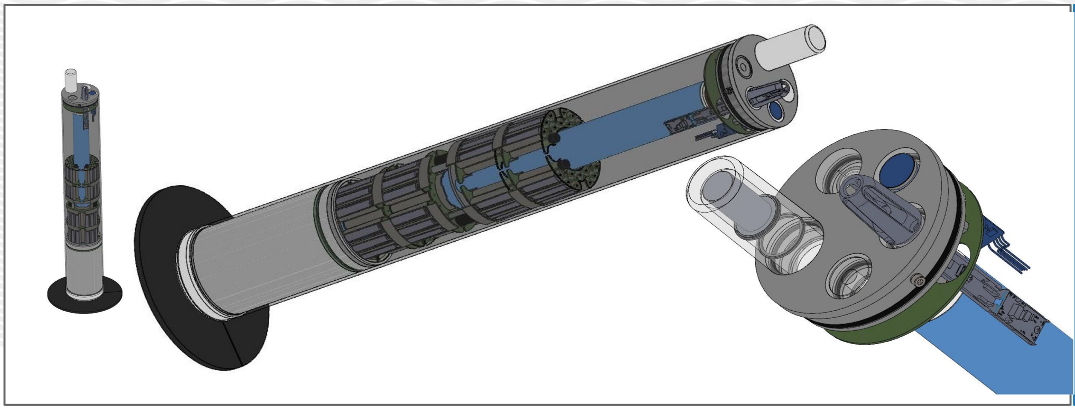
Pacific Marine Environmental Laboratory (PMEL)
pmel.noaa.gov
National Oceanic and Atmospheric Administration
U.S. Department of Commerce

Low-Cost Profiler

www.pmel.noaa.gov/edd/



Overview: Affordable observation profiler designed to be deployed by air, ship or hand for profiling the upper 200m of a water body.

An Affordable Profiler to Enable Ocean Science at Scale

The Low-Cost Profiler (LCP) was conceptualized to reduce the cost of epipelagic oceanographic observations. With a current price point at <$5,000 it is a fraction of the cost of similar floats on the market today. Designed to be deployed by air, ship or hand. The LCP can be configured pre-mission to perform in different observational roles, making it a unique tool for autonomously profiling the upper 200m of a water body. The LCP uses a direct-drive buoyancy engine to reduce mechanical complexity, incorporates OEM manufacturer calibrated sensors, and open-source software (real-time operating system, or RTOS) for simplifying the software complexity.

GENERAL INFORMATION

Cost	<$5,000 (in QTYs of 10 w/ CTD sensor)
Maximum Depth	220m
Endurance	100 profiles over 2 months / 100 profiles over 12 months*
Dimensions	w/o parachute 91.4 cm length x 12. 4 cm diameter w/ parachute 104 cm length x 12.4 cm diameter
Weight	13.6 kg
Hull	Fiberglass
Communication	Iridium Short Burst Data (SBD) w/ more options available in future
GPS Receiver	U-blox ZOE-M8Q
Energy	Alkaline Battery Pack / Rechargeable Lithium*
Features	Free drifting, Moorable, Air deployable*
Sensors	Pressure: Keller 9LX-30Bar (0.3 msw) Temperature: S9 OEM Temp (0.005 deg. C) Conductivity*: Aanderaa 5819 (0.05 PSU), S9 OEM CT (0.05PSU)**, PMEL OEM (0.5 PSU)** CHLA**: Turner Cyclops 7F PAR**: TAG-PARQ Dissolved Oxygen**: Aanderaa optode 5730, PME MicroDOT
* Funded 2026 development; ** Future development dependent on securing funding	

For more information contact: Engineering Development Division

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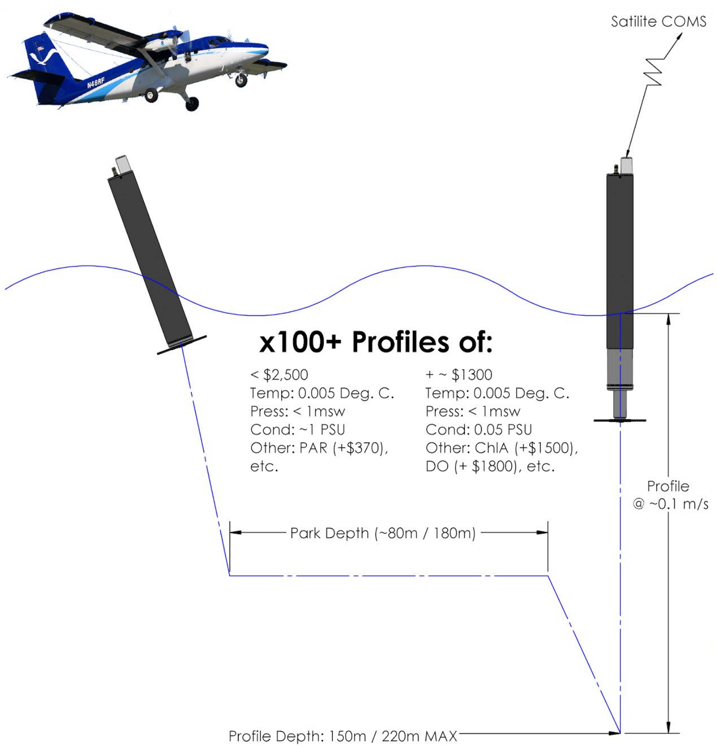
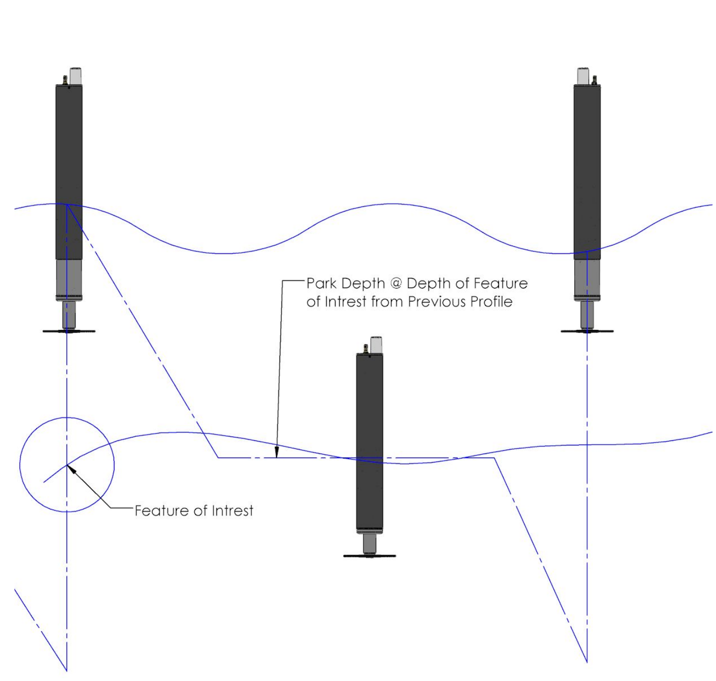
BASIC PROFILE MODE



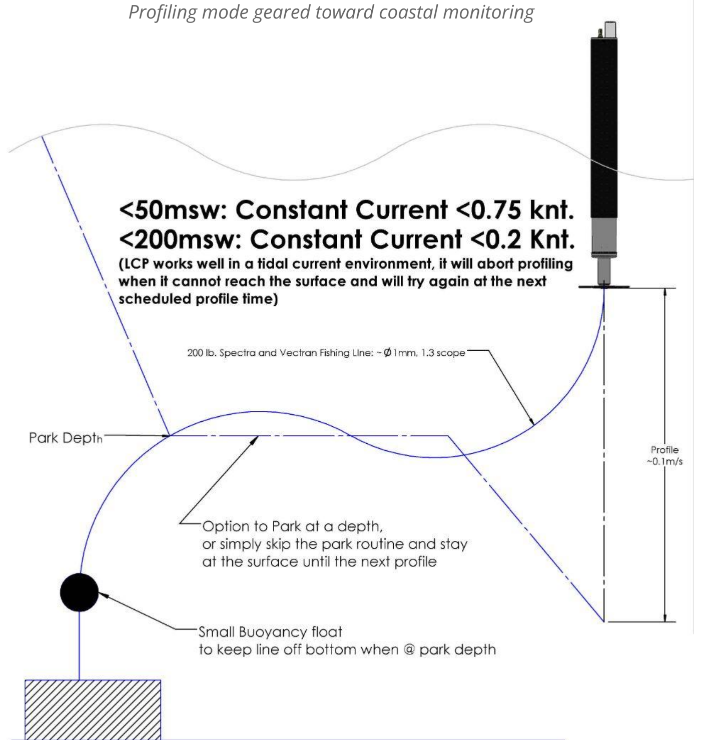
TRACKING PROFILE MODE

Tracking profiling mode to focus a profile on a feature of interest



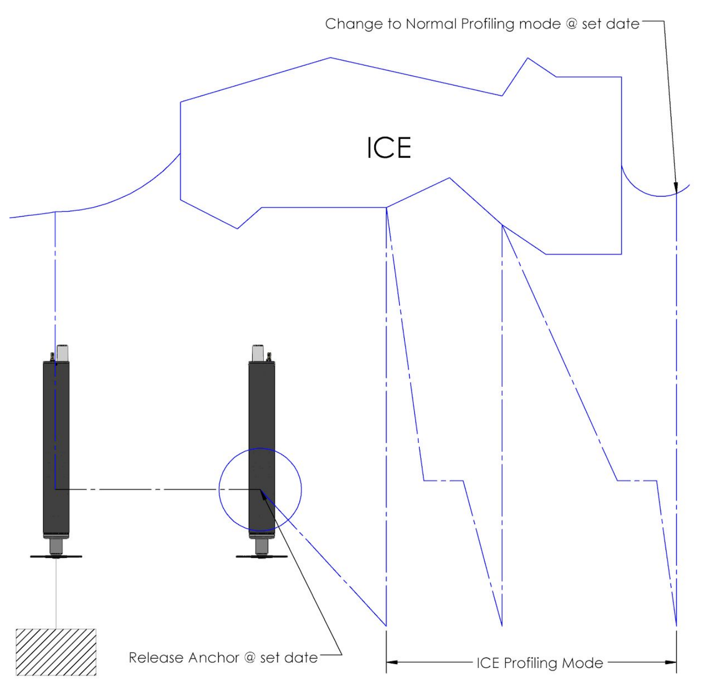
MOORED PROFILE MODE

Profiling mode geared toward coastal monitoring



POPUP PROFILE MODE

Profiling mode geared toward polar monitoring



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